AI Retrieval Architecture

Document RAG Agent

Interactive QA System for Multi-Format Documents powered by LangChain & Streamlit

OpenAI API | LangChain Framework | FAISS Vector Index | Streamlit Interface

System Architecture & Workflow

01

Authentication Layer

Secure API Key gate utilizing Streamlit session state management for session persistence.

02

Document Processing

Dynamic multi-format loading, deep text extraction, and character-level document splitting.

03

Vector Storage

Dense vector embeddings generated via OpenAI models and indexed in high-speed FAISS store.

04

Retrieval & Generation

Context-aware RAG pipeline powered by LangChain chains and OpenAI chat models.

Multi-Format Document Processing

Supported Extensions:

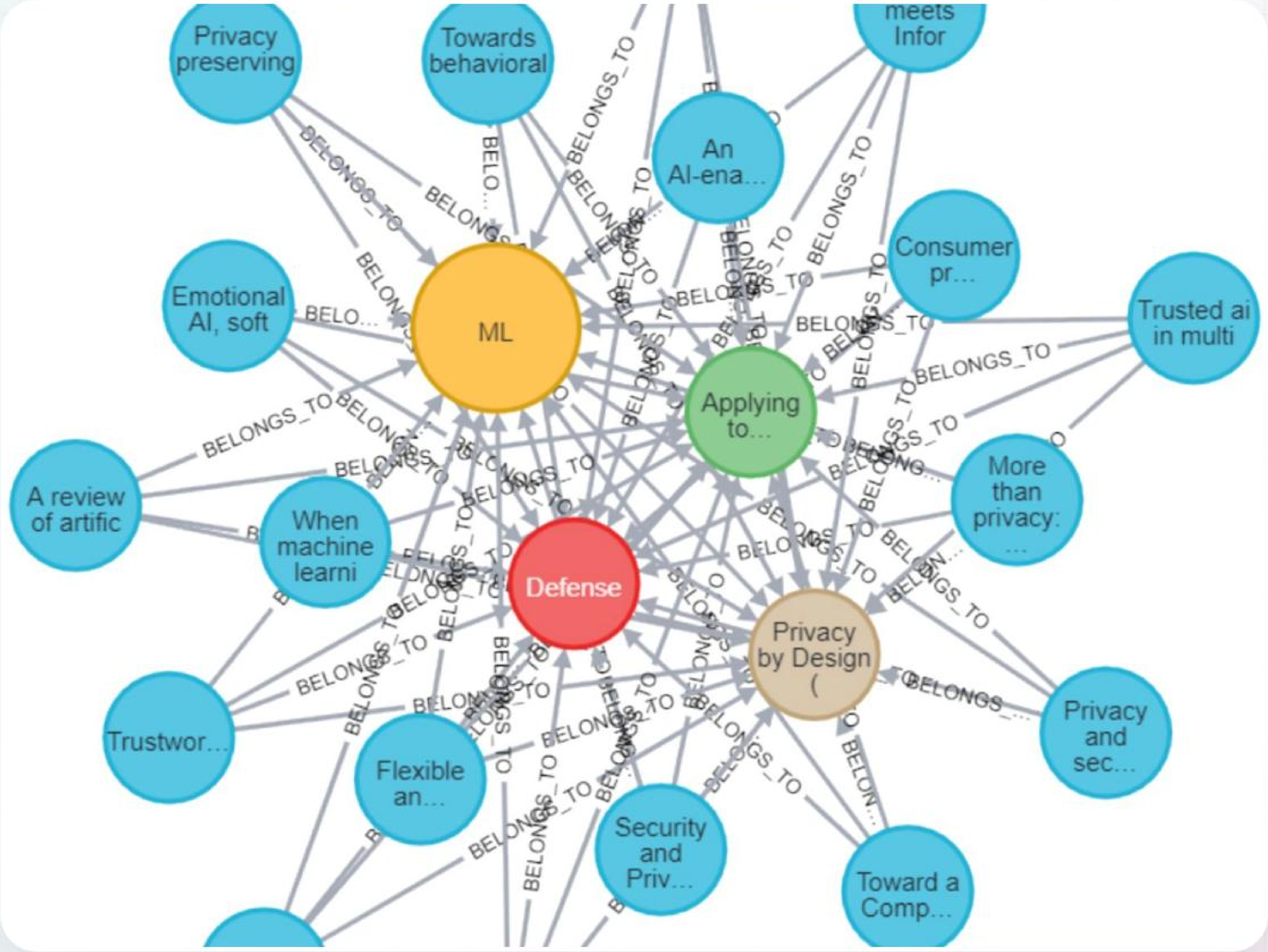
`.pdf` `.docx` `.xlsx` `.pptx` `.ipynb` `.py` `.txt`

Slide & Notebook Parsing

Extracts structured slide text, embedded tables, speaker notes, code cells, and execution outputs.

Robust Text Fallbacks

Multi-encoding decoder handling `utf-8`, `utf-16`, `cp1252`, and `latin-1` gracefully.



Vector Indexing & RAG Chain

Text Chunking

Configurable chunk size and overlap parameters powered by `RecursiveCharacterTextSplitter` to preserve semantic context across chunk boundaries.

Vector Store

High-dimensionality embeddings via `text-embedding-3-small` stored in an ultra-fast, in-memory FAISS vector index.

Prompt Engineering

Strictly constrained system prompt prohibiting hallucination, forcing context-only answers with exact page/source metadata citations.

Streamlit UI & Feature Set

UI/UX Customization

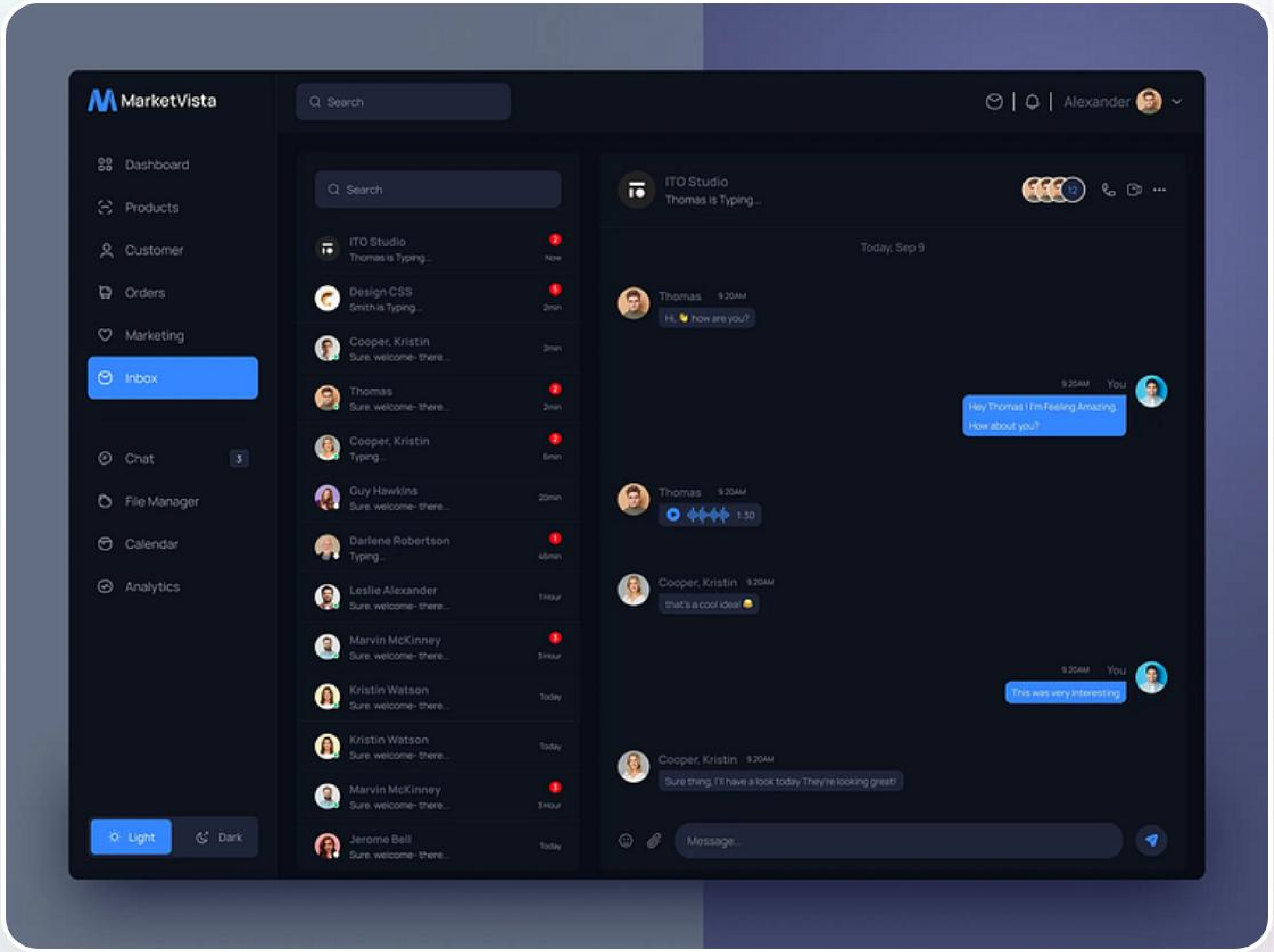
Custom CSS styling with clean typography, elevated cards, and responsive custom control elements.

Interactive Sidebar Controls

Real-time model switching (`gpt-4o-mini`, `gpt-4o`), chunk size sliders, and top-k retrieval depth controls.

Transparent Chat Interface

Streaming LLM response output with expandable source inspection accordions for full document citation trace.



Dependencies & Environment

Core Frameworks

- $ streamlit
- $ langchain
- $ langchain-community
- $ langchain-core

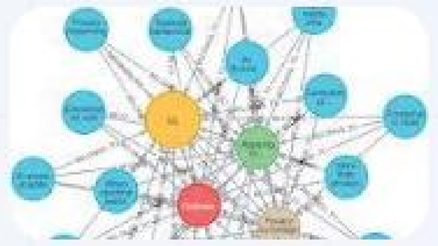
Vector & LLM

- $ langchain-openai
- $ faiss-cpu

Document Parsers

- $ pypdf
- $ python-docx
- $ docx2txt
- $ openpyxl
- $ pandas

Image Sources



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